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APUL

NEET FLIX

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PHYSICS

1) Two particles are executing SHM of the same amplitude A and frequency ω along the x-axis. Their mean position is separated by distance x_0(x_0 > A). If the maximum separation between them is (x_0 + A) the phase difference between their motion is :-

(1) π/2

(2) π/3

(3) π/4

(4) π/6

2)

	Column I		Column II
(a)	Motion of electron around nucleus (as per Bohr's model)	(i)	Oscillatory
(b)	Motion of block connected to spring on smooth horizontal surface	(ii)	Periodic
(c)	Motion of simple pendulum in air medium	(iii)	Both periodic & oscillatory

Match the column

(1) a – (i), b – (ii), c – (iii)

(2) a – (ii), b – (i), c – (iii)

(3) a – (ii), b – (iii), c – (i)

(4) a – (iii), b – (ii), c – (iii)

3) A particle performs simple harmonic motion with amplitude A. Its speed is trebled at the instant that it is at distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is .

(1) 3A

(2) $A\sqrt{3}$

(3) $\frac{7A}{3}$

(4) $\frac{A}{3}\sqrt{41}$

4) For equation y = A sin $\left(\omega t + \frac{\pi}{3}\right)$ match list-I with list-II.

List-I		List - II	
(P)	Motion	(1)	is periodic but not SHM

(Q)	Amplitude	(2)	is SHM
(R)	Initial phase	(3)	A
(S)	Maximum velocity	(4)	π/3
		(5)	None

(1) P→2; Q→3, R→4; S→5

(2) P→5; Q→3, R→4; S→2

(3) P→2; Q→4, R→3; S→5

(4) P→5; Q→3, R→2; S→4

5) A particle executes simple harmonic motion according to equation $4\frac{d^2x}{dt^2} + 320x = 0$. Its time period of oscillation is :-

(1) $\frac{2\pi}{5\sqrt{3}}$ s

(2) $\frac{\pi}{3\sqrt{2}}$ s

(3) $\frac{\pi}{2\sqrt{5}}$ s

(4) $\frac{2\pi}{\sqrt{3}}$ s

6) The time period of a pendulum in an artificial satellite will be

(1) $T = 2\pi\sqrt{\frac{\ell}{g}}$

(2) T = 0

(3) $T \to \infty$

(4) $T = 2\pi\sqrt{\frac{2\ell}{g}}$

7) Average KE in one complete cycle of SHM will be :-

(1) 0

(2) $\frac{1}{4}kA^2$

(3) $\frac{1}{2}kA^2$

(4) $\frac{1}{8}kA^2$

8) A body executes SHM according to equation $x = 10\cos(2\pi t + \pi/4)$. At t = 3/2 s, what is speed of the particle?

(1) 10 ms^{-1}

(2) 20 ms^{-1}

(3) 22 ms–1

(4) 44 ms^{-1}

9) For a particle executing simple harmonic motion, which of the following statements is not correct

(1) The total energy of the particle always remains the same

(2) The restoring force is always directed towards a fixed point

(3) The restoring force is maximum at the extreme positions

(4) The acceleration of the particle is maximum at the equilibrium position

10) A particle executes simple harmonic oscillation with an amplitude a. The period of oscillation is T. The minimum time taken by the particle to travel half of the amplitude from the equilibrium position is :-

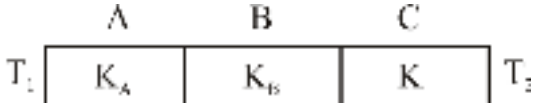
(1) $\frac{T}{4}$

(2) $\frac{T}{8}$

(3) $\frac{T}{12}$

(4) $\frac{T}{2}$

11) Three identical metal rods in dimension A, B and C are placed end to end and a temperature difference is maintained between the ends of A and C. If thermal conductivity of B is thrice that of C and half that of A, then effective thermal conductivity of the system is :-



(1) 12K/7

(2) 6K/7

(3) 7K/3

(4) 2K

12) Find out amount of heat required to convert 2 gm ice at –10°C into water at 60°C.

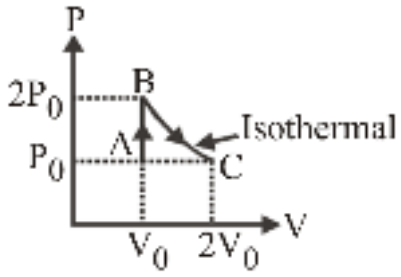
(1) 130 cal

(2) 290 cal

(3) 300 cal

(4) 400 cal

13) P–V curve of a diatomic gas is shown in the figure. Find the total heat given to the gas in the



process A → B → C.

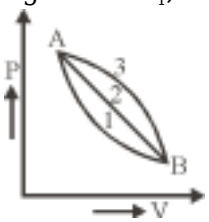
(1) $P_0V_0 + 2P_0V_0\ell n2$

(2) $\frac{1}{2}P_0V_0 + P_0V_0\ell n2$

(3) $\frac{5}{2}P_0V_0 + 2P_0V_0\ell n2$

(4) $3P_0V_0 + 2P_0V_0\ell n2$

14) A given mass of a gas expands from state A to the state B by three paths 1, 2 and 3 as shown in figure. If W_1, W_2 and W_3 respectively be the work done by the gas along the three paths then :-



(1) $W_1 > W_2 > W_3$

(2) $W_1 < W_2 < W_3$

(3) $W_1 = W_2 = W_3$

(4) $W_1 = W_2 < W_3$

15) Internal energy of n_1 moles of hydrogen at temperature T is equal to the internal energy of n_2 mole of helium at temperature 2T. Then the ratio n_1/n_2 is:-

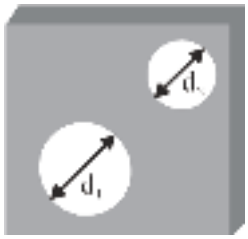
(1) $\frac{3}{5}$

(2) $\frac{2}{3}$

(3) $\frac{6}{5}$

(4) $\frac{3}{7}$

16) Two holes of unequal diameters d_1 and d_2 ($d_1 > d_2$) are cut in a metal sheet. If the sheet is heated



(1) both d_1 and d_2 will decrease

(2) both d_1 and d_2 will increase

(3) d_1 will increase, d_2 will decrease

(4) d_1 will decrease, d_2 will increase

17) An ideal refrigerator has a freezer at a temperature of –13°C. The coefficient of performance of the engine is 5. The temperature of the air (to which heat is rejected) will be :-

(1) 312°C

(2) 325 K

(3) 39°C

(4) 320°C

18) One mole of an ideal monoatomic gas at temperature T^0 expands slowly according to the law $\frac{P}{V}$ = constant. If the final temperature is $2T_0$, heat supplied to the gas is :

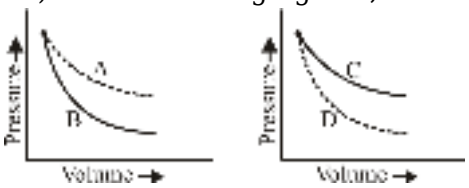
(1) $2RT_0$

(2) RT_0

(3) $\frac{3}{2}RT_0$

(4) $\frac{1}{2}RT_0$

19) In the following figures, four curves A, B, C, D are shown the curves are :–



(1) Isothermal for A and B while adiabatic for C and D

(2) Isothermal for A and C while adiabatic for B and D

(3) Isothermal for A and D while adiabatic for B and C

(4) Adiabatic for A and C while isothermal for B and D

20) Steam at 100ºC is added slowly to 1400 gm of water at 16ºC until the temperature of water is raised to 80ºC. The mass of steam required :-

(1) 120 gm

(2) 240 gm

(3) 160 gm

(4) 320 gm

21) A black body emits 10 watt per cm^2 at 427°C. The sun radiates 10^5 watt per cm^2. Then what is the temperature of the sun?

(1) 5000 K

(2) 6000 K

(3) 7000 K

(4) 8000 K

22) In a process, the molar heat capacity of a diatomic gas is $\frac{10}{3}R$. When heat Q is supplied to the gas, find the work done by the gas :-

(1) $\frac{3Q}{4}$

(2) $\frac{Q}{4}$

(3) $\frac{Q}{3}$

(4) $\frac{2Q}{3}$

23) Oxygen boils at –183°C. This temperature is approximately in Fahrenheit is :-

(1) –329°F

(2) –261°F

(3) –215°F

(4) –297°F

24) A pendulum clock is 5 sec fast at temperature of 15°C and 10 seconds slow at a temperature of 30°C . At what temperature does it give the correct time ?

(1) 18°C

(2) 20°C

(3) 22°C

(4) 25°C

25) An ideal gas at 27°C is compressed adiabatically to $\frac{8}{27}$ of its original volume. If $\gamma = \frac{5}{3}$, then the rise in temperature is :-

(1) 450 K

(2) 375 K

(3) 225 K

(4) 405 K

26)

Temperature of water at the surface of lake is -20^0C. Then temperature of water just below the lower surface of ice layer is-

(1) -4^0C

(2) 0^0C

(3) 4^0C

(4) -20^0C

27) For a gas undergoing an adiabatic change, the relation between temperature and volume is found to be $TV^{0.4}$ = constant. This gas must be

(1) Helium

(2) Carbon dioxide

(3) Argon

(4) Hydrogen

28) An ideal gas whose adiabatic exponent equals to $\gamma = \frac{7}{5}$ is expanded according to the law P = 2V. The initial volume of the gas is equal to V_0 = 1 unit. As a result of expansion the volume increases 4 times. (Take R = $\frac{25}{3}$ units)

Column-I		**Column-II**	
(A)	Work done by the gas	(P)	25 units
(B)	Increment in internal energy of the gas	(Q)	45 units
(C)	Heat supplied to the gas	(R)	75 units
(D)	Molar heat capacity of the gas in the process	(S)	15 units
		(T)	30 units

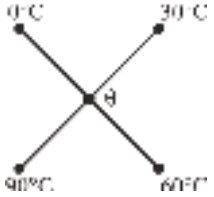
(1) A-S, B-R, C-Q,D-P

(2) A-R, B-Q, C-S,D-P

(3) A-P, B-R, C-S,D-P

(4) A-Q, B-S, C-P,D-R

29) Four rods of same material and having the same cross section and length have been joined, as shown. The temperature of the junction of four rods will be :



(1) 20°C

(2) 30°C

(3) 45°C

(4) 60°C

30) Hot water cools from 60°C to 50°C in the first 10 min and to 42°C in the next 10 min. The temperature of the surrounding is–

(1) 50°C

(2) 10°C

(3) 15°C

(4) 20°C

31) The ratio of coefficient of thermal conductivity of two different materials is 3 : 2. If the thermal resistance of rods of same area of these materials is same, then what is the ratio of length of these rods :

(1) 2 : 3

(2) 3 : 2

(3) 4 : 9

(4) 9 : 4

32) Speed of 5 molecules of a gas are 2, 3, 4, 5, 6 m/sec. The RMS speed of these speeds is :-

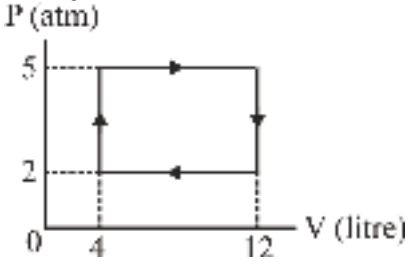
(1) 2.91 m/sec.

(2) 3.52 m/sec.

(3) 4.00 m/sec.

(4) 4.24 m/sec.

33) An ideal gas undergoes a cyclic process as shown in diagram. The net work done by the gas in the cycle is :-



(1) 12 litre-atm

(2) 24 J

(3) 24 litre-atm

(4) –24 J

34) If two rods of length L and 2L (having coefficients of linear expansion α and 2α respectively) are connected so that total length becomes 3L, the average coefficient of linear expansion of the composition rod equals:

(1) $\frac{3}{2}\alpha$

(2) $\frac{5}{2}\alpha$

(3) $\frac{5}{3}\alpha$

(4) $\frac{2}{3}\alpha$

35)

2 kg ice at – 20°C is mixed with 5 kg water at 20°C. Then final amount of water in the mixture would be;
Given specific heat of ice = 0.5 cal/g°C,
Specific heat of water = 1 cal/g°C,
Latent heat of fusion for ice = 80 cal/g.

(1) 6 kg
(2) 5 kg
(3) 4 kg
(4) 2 kg

36) An ideal gas heat engine operates in Carnot cycle between 227°C and 127°C. It absorbs 6×10^4 cals of heat at higher temperature. Amount of heat converted to work is

(1) 2.4×10^4 cal
(2) 6×10^4 cal
(3) 1.2×10^4 cal
(4) 4.8×10^4 cal

37) A carnot engine has the same efficiency between 800 K to 500 K and x K to 600 K. The value of x is

(1) 1000 K
(2) 960 K
(3) 846 K
(4) 754 K

38) Cooling rate of a sphere of 600 K at external environment (200 K) is R. When the temperature of sphere is reduced to 400 K then cooling rate of the sphere becomes :

(1) $\frac{3}{16}R$
(2) $\frac{16}{3}R$
(3) $\frac{9}{27}R$
(4) None

39) The rate of emission of radiation of a black body at 273°C is E, then the rate of emission of radiation of this body at 0°C will be -

(1) E/16
(2) E/4
(3) E/8
(4) 0

40) There is a black spot on a body. If the body is heated and carried in a dark room then it glows

more. This can be explained on the basis of -

(1) Newton's law of cooling
(2) Wein's law
(3) Kirchhoff's law
(4) Stefan's law

41) If the root mean square speed of hydrogen molecules is equal to root mean square speed of oxygen molecules at 470C, the temperature of hydrogen is -

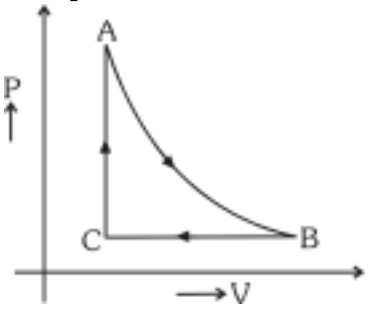
(1) 20 K
(2) 47 K
(3) 50 K
(4) 94 K

42) ***Assertion :*** The value of ΔQ is always zero in adiabatic process.
Reason : Adiabatic process is always a cyclic process.

(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are False.

43) Which of the following is a state function :-

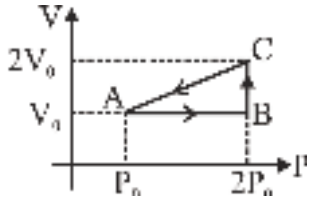
(1) Work done in cyclic process
(2) Work done in isothermal process
(3) Heat at constant pressure
(4) Heat at constant volume

44) For one mole of an ideal gas at 300 K occupies a volume of 0.36 m^3 at a pressure of 2 atm. The gas expands adiabatically until its volume becomes 1.44 m^3. Next gas is compressed isobarically to its original volume. Finally, the pressure is increased isochorically until the gas returns to its initial state. (see figure). ($\gamma = \frac{3}{2}$) 

Determine the temperature at the point B

(1) 150 K
(2) 200 K
(3) 400 K
(4) 300 K

45) A thermodynamic process of one mole ideal monoatomic gas is shown in figure. The efficiency of cyclic process ABCA will be :-



(1) 25%

(2) 12.5%

(3) 50%

(4) $\frac{100}{13}\%$

CHEMISTRY

1) At 400K, 1 mol of a hydrocarbon is completely burnt. It gives 132 g of CO_2 along with 72 g of water vapour then hydrocarbon may be :-

(1) CH_4

(2) C_3H_8

(3) C_2H_4

(4) C_4H_{10}

2) The ratio of mass of 1g atom carbon and 1g molecule oxygen will be :-

(1) 1 : 1

(2) 12 : 16

(3) 12 : 32

(4) 12 : 8

3) For the reaction $2P + Q \rightarrow R$, 8 moles of P and 5 moles of Q will produce :

(1) 8 mol of R

(2) 5 mol of R

(3) 4 mol of R

(4) 13 mol of R

4) How many electrons are present in 1.6g of CH_4 ?

(1) 6.02×10^{23} electrons

(2) 6.02×10^{22} electrons

(3) 6.02×10^{24} electrons

(4) 6.02×10^{25} electrons

5) Actual mass of He-atom is :-

(1) 4g

(2) $4 \times 6.02 \times 10^{23}$ g

(3) $4 \times 1.67 \times 10^{-24}$ g

(4) $2 \times 1.67 \times 10^{-24}$ g

6) The empirical formula of a compound is CH. Its molecular weight is 78. The molecular formula of the compound will be :

(1) C_2H_2

(2) C_3H_3

(3) C_2H_4O

(4) C_6H_6

7) Haemoglobin contains 0.334% of iron by weight. The molecular weight of haemoglobin is approximately 67200. The number of iron atoms (atom is weight of iron is 56) present in one molecule of haemoglobin is

(1) 4

(2) 6

(3) 3

(4) 2

8) The number of oxygen atoms in 6.4 g of SO_2 is :-

(1) 6×10^{23}

(2) 11×10^{23}

(3) 12×10^{23}

(4) 1.2×10^{23}

9) When 22.4 L of H_2 is mixed with11.2 L of Cl_2(g) at STP the moles of HCL formed is equal to:

(1) 1 mole of HCl (g)

(2) 2 moles of HCl (g)

(3) 0.5 moles of HCl (g)

(4) 1.5 moles of HCl (g)

10) How much volume will be occupied by 2 mole CO_2 gas at STP?

(1) 22.4 L

(2) 44.8 L

(3) 22.7 L

(4) 45.4 L

11) In which case is number of moles of water maximum ?

(1) 18 mL of water

(2) 0.18 g of water

(3) 10^{-3} g of water

(4) 0.00224 L of water vapours at 1 atm and 273 K

12) A mixture of gases contains H_2 and O_2 gases in the ratio of 1:4 (w/w). What is the molar ratio of the two gases in the mixture ?

(1) 2 : 1

(2) 1 : 4

(3) 4 : 1

(4) 16 : 1

13) A compound possesses 8% sulphur by mass. The least molecular mass is :

(1) 200

(2) 400

(3) 155

(4) 355

14) 1 mole of oxygen gas occupy 22.4 L at STP. Its density will be:

(1) 1.43 g/L

(2) 0.98 g/L

(3) 1.14 g/L

(4) 0.58 g/L

15) A gas is found to have a formula $(CO)_x$. It's vapour density is 140 the value of x must be

(1) 4

(2) 8

(3) 5

(4) 10

16) An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide will be :-

(1) M_2O

(2) M_2O_3

(3) MO

(4) M_3O_4

17) 1 L of a hydrocarbon weighs as much as one litre of CO_2 Under similar conditions. Then the molecular formula of the hydrocarbon is -

(1) C_3H_8

(2) C_2H_6

(3) C_2H_4

(4) C_3H_6

18) 18 gms of Al will react, with $2Al + \frac{3}{2} O_2 \rightarrow Al_2O_3$

(1) 16 gms O_2

(2) 8 gms O_2

(3) 9 gms O_2

(4) 4 gms O_2

19) 25 gm of $CaCO_3$ is thermally decomposed. The volume of CO_2 obtained at STP is :-

(1) 11.2 lit.

(2) 22.4 lit.

(3) 2.8 lit.

(4) 5.6 lit.

20) 26 cc of CO_2 are passed over red hot coke. The volume of CO evolved is :-

(1) 15 cc

(2) 10 cc

(3) 32 cc

(4) 52 cc

21) A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. The value of K_p if the total pressure at equilibrium is 0.8 atm, is

(1) 1.8 atm

(2) 3 atm

(3) 0.3 atm

(4) 0.18 atm

22)

The equation:

$$2Al(s) + \frac{3}{2} O_2(g) \rightarrow Al_2O_3(s)$$

shows that:-

(1) 2 mole of Al reacts with 3/2 mole of O_2 to produce 7/2 mole of Al_2O_3

(2) 2g of Al reacts with 3/2 g of O_2 to produce one mole of Al_2O_3

(3) 2 g mole of Al reacts with 3/2 mole of O_2 to produce one mole of Al_2O_3

(4) 2 mole of Al reacts with 3/2 mole of O_2 to produce 1 mole of Al_2O_3

23) **Assertion :** The equivalent mass of an element is variable.
Reason : It depends on the valency of the element.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

24)

If law of conservation of mass was to hold true, then 20.8 gm. of $BaCl_2$ on reaction with 9.8 gm. of H_2SO_4 will produce 7.3 gm. of HCl and $BaSO_4$ equal to -

(1) 11.65 gm
(2) 23.3 gm
(3) 25.5 gm
(4) 30.6 gm

25) **Assertion:** Both 138 g of K_2CO_3 and 12 g of carbon have same number of carbon atoms.
Reason: Both contain 1 g mole of carbon which contains 6.022×10^{23} carbon atoms.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(3) Assertion is correct, but Reason is incorrect.
(4) Both Assertion and Reason are incorrect.

26) Which of the following follows the law of multiple proportion?

(1) CO, CO_2
(2) NH_3, PH_3
(3) $CaCl_2$, $MgCl_2$
(4) H_2S, CS_2

27) Which of the following contains the largest mass of hydrogen atoms ?

(1) 5.0 moles $C_2H_2O_4$
(2) 1.1 moles $C_3H_8O_3$
(3) 1.5 moles $C_6H_8O_6$
(4) 4.0 moles $C_2H_4O_2$

28) What is the vapour density of O_3 with respect to CH_4

(1) 3
(2) 24
(3) 48
(4) 6

29) Elements A and B form two compounds B_2A_3 and B_2A. 0.05 mole of B_2A_3 has mass 9 g and 0.1 mole B_2A has mass 10 g. Find atomic weight of A & B respectively.

(1) 20 and 30
(2) 30 and 40
(3) 40 and 30
(4) 30 and 20

30) Which of the following compounds has same empirical formula as that of glucose-

(1) CH_3CHO
(2) CH_3COOH
(3) CH_3OH
(4) C_2H_6

31) Based on the following reaction which one oxide is more stable :-
$2MO_{(g)} \rightleftharpoons M_{2(g)} + O_{2(g)} K_1 = 1 \times 10^{-10}$
$2MO_{2(g)} \rightleftharpoons M_{2(g)} + 2O_{2(g)} K_2 = 4 \times 10^{6}$

(1) MO
(2) MO_2
(3) Both (1) and (2)
(4) None

32) For which of the following reactions $K_p < K_c$:-

(1) $N_2O_3(g) \rightleftharpoons NO_2(g) + NO(g)$
(2) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$
(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(4) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

33) What is equilibrium expression for the reaction $P_{4(s)} + 5O_{2(g)} \rightleftharpoons P_4O_{10(s)}$:-

(1) $K_c = \frac{[P_4O_{10}]}{[P_4][O_2]^5}$
(2) $K_c = \frac{[P_4O_{10}]}{5[P_4][O_2]^5}$
(3) $K_c = [O_2]^5$
(4) $K_c = \frac{1}{[O_2]^5}$

34) In which of the following does the reaction go farthest to completion?

(1) $K = 10^3$
(2) $K = 10^{-2}$

(3) K = 10

(4) K = 1

35) On increasing temperature, the equilibrium constant of exothermic and endothermic reactions respectively:-

(1) Increases and Decreases

(2) Decreases and Increases

(3) Increases and Increases

(4) Decreases and Decreases

36) At 500 K, equilibrium constant, K_c, for the following reaction is 5.

$$\frac{1}{2}H_2(g)+\frac{1}{2}I_2(g)\rightleftharpoons HI(g)$$

What would be the equilibrium constant K_c for the reaction, $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$?

(1) 0.04

(2) 0.4

(3) 25

(4) 2.5

37) 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium :

(1) 4.5

(2) 6

(3) 3

(4) 4

38) Effect of increasing temperature on equilibrium constant is given by $\log K_2 - \log K_1 = \frac{-\Delta H}{2.303R}\left[\frac{1}{T_2}-\frac{1}{T_1}\right]$ Then for an endothermic reaction the false statement is:-
[where T_2 - final temperature, T_1 - initial temperature].

(1) $\left[\frac{1}{T_2}-\frac{1}{T_1}\right]$ = positive

(2) $\log K_2 > \log K_1$

(3) ΔH = positive

(4) $K_2 > K_1$

39) For the reaction $A(g) + 3B(g) \rightleftharpoons 2C(g)$ at 27°C, 2 moles of A, 4 moles of B and 6 moles of C are present in 2 litre vessel. If K_c for the reaction is 1.2, the reaction will proceed in :-

(1) Forward direction

(2) Backward direction

(3) Neither direction

(4) None of these

40) At constant pressure the addition of argon to $N_2+3H_2 \rightleftharpoons 2NH_3$ will

(1) Shift the equilibrium in forward direction

(2) Shift the equilibrium in backward direction

(3) Not affect the equilibrium

(4) Stop the reaction

41) Some inert gas is added at constant volume of the following reaction at equilibrium $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$
Predict the effect of adding the inert gas :-

(1) The equilibrium shifts in the forward direction

(2) The equilibrium shifts in the backward direction

(3) The equilibrium remains unaffected

(4) The value of K_p is increased

42) The vapour density of undecomposed N_2O_4 is 46, now on heating at high temperature. The vapour density of equilibrium mixture is found to be 30. The percentage dissociation of N_2O_4 at equilibrium for the reaction given below is :- $N_2O_4 \rightleftharpoons 2NO_2$

(1) 33.3%

(2) 11.1%

(3) 53.3%

(4) 83.3%

43) In the following equilibria,
I. $A + 2B \rightleftharpoons C$; $K_{eq} = K_1$
II. $C + D \rightleftharpoons 3A$; $K_{eq} = K_2$
III. $6B + D \rightleftharpoons 2C$; $K_{eq} = K_3$
Hence,

(1) $3K_1 \times K_2 = K_3$

(2) $K_1^3 \times K_2^2 = K_3$

(3) $3K_1 \times K_2^2 = K_3$

(4) $K_1^3 \times K_2 = K_3$

44) For an equilibrium, $H_2O(s) \rightleftharpoons H_2O(\ell)$ which of the following statement is true :-

(1) The pressure changes do not affect the equilibrium

(2) More of ice melts if pressure on the system is increased

(3) More of liquid freezes if pressure on the system is increased

(4) None of these

45) **Assertion (A) :** Addition of an inert gas at constant volume to the equilibrium mixture has no effect on the state of equilibrium.

Reason (R) : The addition of inert gas at constant volume will not affect the concentration or $\left(\frac{n}{V}\right)$ of the reactants as well as products of a reaction mixture.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true and R is not a correct explanation of A.

(3) A is true, but R is false

(4) A is false, but R is true.

BIOLOGY

1) Choose correct answer:
(A) Nuclear membrane is absent in protista
(B) Animalia are consumer in nature
(C) Cell wall is noncellulosic in Monera
(D) Protista is a prokaryotic organism

(1) A

(2) B and C

(3) C and D

(4) A and D

2) Which of the following is a parasitic fungi on wheat ?

(1) *Rhizopus*

(2) *Puccinia*

(3) *Albugo*

(4) *Neurospora*

3) Which one of the following structure is similar in *E.coli* and *Chlamydomonas* ?

(1) Cell wall

(2) Cell membrane

(3) Golgi body

(4) Chromosomal Organization

4) Choose the correct match

(1) Chrysophytes - Walls are indestructible

(2) Dinoflagellates - Lack cell wall

(3) Euglenoides - Unicellular prokaryote

(4) Slime moulds - Chitinous cell wall

5) Identify the correct match and select the correct option from options given below :
(A) Yeast : Unicellular fungi : Basidiomycetes

(B) *Albugo* : Parasitic fungi : Phycomycetes
(C) *Puccinia* : Rust fungi : Basidiomycetes
(D) Truffles : Edible mushroom : Ascomycetes

(1) A, B, C and D
(2) B and C
(3) B, C and D
(4) A, B and C

6) Which one is incorrect statement ?

(1) Protista have photosynthetic and heterotrophic modes of nutrition.
(2) Some fungi are edible.
(3) Nuclear membrane is present in kingdom Monera.
(4) Cell wall is absent in Animalia.

7) Match the following :-

	Column-I		**Column-II**
(A)	Diatom	(i)	Mixotrophic nutrition
(B)	Dinoflagellates	(ii)	Plasmodium
(C)	Euglenoids	(iii)	Diatomaceous earth
(D)	Slime moulds	(iv)	Plated cell wall

(1) A-iii, B-i, C-iv, D-ii
(2) A-iii, B-iv, C-i, D-ii
(3) A-iii, B-iv, C-ii, D-i
(4) A-iv, B-iii, C-i, D-ii

8) Imagine an organism found having flagella, absorptive mode of nutritional, photosynthetic, and show prokaryotic organisation. Which kingdom does the organism belong ?

(1) Protista
(2) Plantae
(3) Monera
(4) Animalia

9) Choose the correct statement.

(1) Chrysophytes are plankton which floats actively in water current.
(2) Dinoflagellates have two flagella which lies longitudinally.
(3) Pigments of euglenoids are similar to higher plant.
(4) Slime mould posses true walls in plasmodium.

10) How many of the following are sexual spores in fungi ?

Conidia, sporangiospore, zoospore, ascospore, basidiospore, oospores

(1) 1
(2) 2
(3) 3
(4) 4

11) Choose the incorrect match -

	I	II
(1)	Chief producer of ocean	Chrysophytes
(2)	Plasmodium formation	Slime mould
(3)	Red tides	Euglenoid
(4)	Stiff cellulose plates	Dinoflagellate

(1) 1
(2) 2
(3) 3
(4) 4

12) In two kingdom classification. The main character of the members of plantae was :-

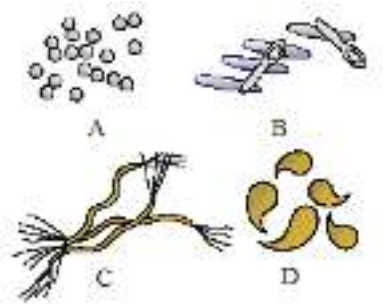
(1) Autotrophic in nature
(2) Presence of cell wall
(3) Starch as reserve food
(4) Non motile organisms

13) The classification system have undergone several changes over a period of time, why ?

(1) Because the criteria for classification changed, now they are based to trace the developmental history
(2) Because over time an attempt has been made to evolve such system which are based on histological and cytological studies
(3) Because over time an attempt has been made to evolve such classification system which are based only on human evolution mainly
(4) Because over time an attempt has been made to evolve such classification which are based on phylogeny

14) Choose the incorrect statement regarding Mycoplasma :-

(1) They lack cell wall
(2) They are smallest living cell
(3) They can survive without oxygen
(4) They have mitochondria for respiration

15) 

Select option showing correct identification for above figure from A to D :-

	A	B	C	D
(1)	*Bacillus*	*Coccus*	*Spirillum*	*Vibrio*
(2)	*Coccus*	*Bacillus*	*Spirillum*	*Vibrio*
(3)	*Coccus*	*Spirillum*	*Bacillus*	*Vibrio*
(4)	*Coccus*	*Vibrio*	*Bacillus*	*Spirillum*

(1) 1

(2) 2

(3) 3

(4) 4

16) Select out incorrect one with respect to monera

(1) Most of the member are heterotrophic

(2) All moreran have flagella

(3) Some member show symbiotic relation

(4) Some member show aerobic respiration

17)  Identify the given figure :-

(1) Protosynthetic diatom

(2) Euglena

(3) Paramecium

(4) Dinoflagillates

18) In five kingdom classification, the kingdom that includes the blue-green algae, nitrogen-fixing bacteria and methanogenic archaebacteria, is

(1) Monera

(2) Protista

(3) Fungi

(4) Plantae

19) Dikaryon formation is the characteristic feature of :

(1) Ascomycetes and Basidiomycetes.
(2) Phycomycetes and Basidiomycetes.
(3) Ascomycetes and Phycomycetes.
(4) Phycomycetes and Zygomycetes.

20) Under favourable conditions, bacteria undergo __A__ while under unfavourable conditions they form __B__. Complete the above statement by choosing the correct option for A and B respectively :

(1) Fragmentation, Ascospores
(2) Binary fission, Fruiting bodies
(3) Binary Fission, Endospores
(4) Plasmodium, conidia

21) **Statement-I :** Pigments of Euglenoids are identical to green algae.
Statement-II : According to two kingdom classification *Chlorella* and *Chlamydomonas* are included in kingdom Protista.

(1) Only statement-II in correct
(2) Statement-I and II both are correct
(3) Only statement-I is correct
(4) Statement I and II both are incorrect

22) Study the four statements (A–D) carefully :-
(A) Bacteria show most extensive metabolic diversity
(B) Some of the bacteria are autotrophs
(C) Bacteria are the sole members of kingdom Protista
(D) Bacteria also live extreme habitats such as hot springs, desert, snow and deep ocean
How many of the above statements are correct?

(1) One
(2) Two
(3) Three
(4) Four

23) The protein coat of viruses is made up of small subunits called :-

(1) Flagellins
(2) Virions
(3) Pilin
(4) Capsomeres

24) Match the column A with column B and find out the correct option as shown ?

	Column-A		**Column-B**

(a)	Obligate parasite	(p)	ss DNA
(b)	ϕ×174	(q)	Virus
(c)	Potato spindle tuber disease	(r)	Viroids
(d)	Corona virus	(s)	RNA

(1) a-q, b-p, c-r, d-s

(2) a-p, b-s, c-r, d-q

(3) a-p, b-q, c-r, d-s

(4) a-q, b-r, c-s, d-p

25) Read the following statements carefully :-
(A) Members of this kingdom show a great diversity in morphology and habitat
(B) Some members of this kingdom are used to make bread and beer some are the source of antibiotics.
(C) Their bodies consist of long, slender thread like structure called hyphae
(D) They likely to grow in warm and humid places
On the basis of above statements identify the kingdom :-

(1) Protista

(2) Monera

(3) Fungi

(4) Plantae

26) Which kingdom forms a link with the others dealing with plants, animals and fungi.

(1) Monera

(2) Fungi

(3) Protista

(4) Animalia

27) Animal like feature of slime moulds is–

(1) Formation of fruiting bodies

(2) Cellulosic cell wall around spores

(3) Absence of cell wall in plasmodium

(4) Formation of spores

28) Classification of plants based on evolutionary relationship is called :

(1) Artificial system

(2) Natural system

(3) Phylogenetic system

(4) Numerical taxonomy

29) *Chlorella* are placed according to Whittaker classification, in :

(1) Pyrophyta

(2) Protozoans

(3) Protista

(4) Both in pyrophyta and chrysophyta

30) Match the column I, II and III

	Column-I		**Column-II**		**Column-III**
A	Monera	I	Cellulosic cell wall	P	Parasitic and saprophytic
B	Protista	II	Nuclear membrane absent	Q	Holozoic
C	Fungi	III	May form loose tissue	R	Photosynthetic
D	Plantae	IV	Tissue/organ grade of organisation	S	Cell wall made of chitin

(1) A-I,Q ; B-II,P,S ; C-IV,P ; D-III,Q

(2) A-II,P ; B-I,P,Q,R ; C-III,S,P ; D-I,IV,R

(3) A-III,Q ; B-II,P,S ; C-III,S,P ; D-II,Q

(4) A-IV,P ; B-I,R,S ; C-II,Q ; D-III,P

31) *Chlorella, Chlamydomonas* and *Paramoecium, Amoeba* were earlier placed with plants and animals respectively but after Whittaker's 5 kingdom classification, they should be brought together in:-

(1) Monera

(2) Protista

(3) Plantae

(4) Animalia

32) As we go from species to kingdom in a taxonomic hierarchy, the number of common characteristics :-

(1) Will decrease

(2) Will increase

(3) Remain same

(4) May increase or decrease

33) Study the following table which shows different organisms with their taxonomic categories :-

Common name	**Family**	**Order**	**Class**	**Phylum/Division**

Man	Hominidae	Primata	Mammalia	A
Housefly	Muscidae	Diptera	B	Arthropoda
Mango	C	Sapindales	Dicotyledonae	Angiospermae
Wheat	Poaceae	Poales	D	Angiospermae

Select the correct option for A, B, C and D

(1) A–Chordata, B–Insecta,
C–Anacardiaceae, D–Monocotyledonae

(2) A–Animalia, B–Mammalia,
C–Anacardiaceae, D–Dicotyledonae

(3) A–Chordata, B–Mammalia,
C–Polygonaceae, D–Monocotyledonae

(4) A–Non-chordata, B–Insecta,
C–Anacardiaceae, D–Dicotyledonae

34) Which of the following option is correctly matched :-

(1) Man - Diptera

(2) Housefly - Primata

(3) Mango - Sapindales

(4) Wheat - *Musca*

35) A group of related taxons ending with suffix—'ales' will constitute category:

(1) Family

(2) Genus

(3) Order

(4) Class

36) Which of the following statements regarding nomenclature is correct?

(1) Generic name always begins with capital letter whereas specific name with small letter.

(2) Scientific name should be printed in italics.

(3) Scientific name when typed or handwritten should be underlined.

(4) All of the above

37) Read the following statement A and B and choose the correct option :
Statement A : The number of common characteristics goes on increasing from species to kingdom
Statement B : Higher the category, greater is the difficulty of determining the relationship to other taxa of same level.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

38) Plant families like convolvulaceae, solanaceae are included in the order :-

(1) Polymoriales
(2) Sapindales
(3) Mangifera
(4) Anacardiaceae

39) Match the column and choose the correct option-

	Column-I		**Column-II**
(A)	Man	(i)	*Homo sapiens*
(B)	Housefly	(ii)	*Musca domestica*
(C)	Mango	(iii)	*Mangifera indica*
(D)	Wheat	(iv)	*Triticum aestivum*

(1) A-(iv), B-(iii), C-(ii), D-(i)
(2) A-(iv), B-(iii), C-(i), D-(ii)
(3) A-(i), B-(ii), C-(iii), D-(iv)
(4) A-(iii), B-(iv), C-(ii), D-(i)

40) Which of the following is not an essential taxonomic categories?

(1) Kingdom
(2) Domain
(3) Species
(4) Order

41) Families are characterised on the basis of-

(1) Vegetative features only
(2) Reproductive features only
(3) Both the reproductive and vegetative features
(4) Phylogeny only

42) Sapindales, Polymoniales, Carnivora, Primata, Diptera are-

(1) Same taxa at same level
(2) Different taxa at different levels
(3) Different taxa at same level
(4) Taxon

43) **Assertion :** Dinoflagellates are called "Fire algae".
Reason : They appear as glowing light due to bioluminescence.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.
(3) Assertion is correct and reason is incorrect.
(4) Both assertion and reason are incorrect.

44) Largest unit of classification is:

(1) Genus
(2) Species
(3) Order
(4) Kingdom

45) The common characteristics between tiger and cats will be maximum at the level of their :-

(1) Class
(2) Genus
(3) Species
(4) Family

46) Which of the following character is not related with chordates ?

(1) Ventral nerve cord
(2) Ventral heart
(3) Dorsal notochord
(4) Post anal tail

47) Which is not a character of cartilaginous fishes?

(1) 2- Chambered heart
(2) Paired fins
(3) Air or swim bladder
(4) Gills without operculum

48) The characteristic feature of birds which makes them different from reptiles is -

(1) Homoeothermic nature
(2) Post anal tail
(3) Vertebral column
(4) Viviparity

49) Select the correct option :

Animal		**Common name**
(a)	*Gorgonia*	(1) Bath sponge
(b)	*Ctenoplana*	(2) Sea fan
(c)	*Taenia*	(3) Comb jelly

(d)	*Euspongia*	(4) Tapeworm

(1) a-3, b-2, c-4, d-1

(2) a-1, b-3, c-4, d-2

(3) a-2, b-3, c-1, d-4

(4) a-2, b-3, c-4, d-1

50) Diploblastic and triploblastic are terms that describe

(1) The number of invaginations during embryonic development

(2) The number of heads during embryonic development

(3) The number of germinal layers during embryonic development

(4) The number of cell types during development

51) Which of the following statement is correct ?
(i) In urochordates like Ascidia, Salpa, Doliolum, notochord is present only in larval tail.
(ii) In cephalochordates like Branchiostoma (Amphioxus or Lancelet) notochord extends from head to tail region and is persistent throughout the life.
(iii) In chondricthyes mouth is located dorsally and notochord is persistent throughout the life.
(iv) In cyclostomates like Petromyzon scales, Jaw , cranium, Vertebral column are absent

(1) (i), (ii), (iii)

(2) (i) and (ii)

(3) All except (iv)

(4) All of these

52) Fertilisation is internal in :-

(1) *Rana*, *Psittacula*, *Scoliodon*

(2) *Hippocampus*, *Rana*, *Testudo*

(3) *Neophron*, *Chameleon, Scoliodon*

(4) *Bufo*, *Clarias*, *Pavo*

53) Open circulatory system is not present in :-

(1) Tunicates

(2) Mollusca

(3) Arthropoda

(4) Annelida (Earthworm, Nereis)

54) Find out the correctly matched :-

	Column-I	**Column-II**
(1)	Aplysia	Cuttle fish
(2)	Fasciola	Hook worm
(3)	Physalia	Portugese Man of war

(4)	Antedon	Brittle star

(1) 1
(2) 2
(3) 3
(4) 4

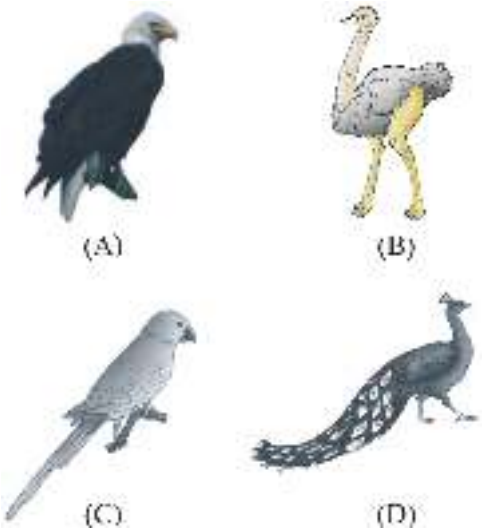
55) Metagenesis means :-

(1) Polyp produce medusae asexually and medusae produce polyp sexually
(2) Polyp produce medusae sexually and medusae produce polyp sexually
(3) Polyp produce medusae asexually and medusae produce polyp asexually.
(4) Both (1) & (2)

56) Claspers are :-

(1) Respiratory and present in reptilia
(2) Present on pelvic fins in cartilaginous fishes
(3) Olfactory and present in amphibia
(4) Respiratory and present in amphibia

57) Identify the name of given animals from the given option.



	A	**B**	**C**	**D**
(1)	*Struthio*	*Neophron*	*Pavo*	*Psittacula*
(2)	*Neophron*	*Struthio*	*Pavo*	*Psittacula*
(3)	*Neophron*	*Struthio*	*Psittacula*	*Pavo*
(4)	*Pavo*	*Struthio*	*Psittacula*	*Neophron*

(1) 1
(2) 2
(3) 3
(4) 4

58) Give the correct match in the following :-

Scientific Name		**Common Name**	
(A)	Balenoptera	(I)	Tortoise
(B)	Pteropus	(II)	Angel fish
(C)	Testudo	(III)	Flying fox
(D)	Pterophyllum	(IV)	Blue whale

	A	**B**	**C**	**D**
(1)	II	III	I	IV
(2)	IV	III	I	II
(3)	IV	I	II	III
(4)	I	II	III	IV

(1) 1

(2) 2

(3) 3

(4) 4

59) The characteristic of class Reptilia are :-

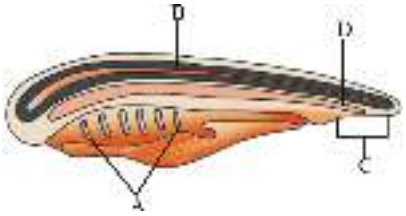
(1) Body covered with moist skin which is devoid of scales, the ear is represented by tympanum, alimentary canal, urinary and reproductive tract open into a common cloaca.

(2) Fresh water, animals with bony endoskeleton, air bladder to regulate buoyancy.

(3) Marine animals with cartilagineous endoskeleton.

(4) Body covered with dry and cornified skin, with epidermal scales or scutes

60) In the figure given below, Identify the parts labelled as A,B,C and D and select correct option about them :-



	Part-A	Part-B	Part-C	Part-D
(1)	Notochord	Nerve Cord	Post Anal tail	Pharyngeal gill slits
(2)	Nerve Cord	Notochord	Pharyngeal gill slits	Post Anal tail
(3)	Pharyngeal gill slits	Nerve Cord	Post Anal tail	Notochord

(4)	Post Anal Tail	Pharyngeal gill slits	Notochord	Nerve Cord

(1) 1

(2) 2

(3) 3

(4) 4

61) Read the following statements with respect to Aves:-
(i) Respiration is through skin.
(ii) The fore limbs always possess scales.
(iii) Long bones are pneumatic.
(iv) Heart is completely three chambered.
(v) Digestive tracts have additional chambers called crop and gizzard.
Which of the above statements are correct?

(1) (iii), (v)

(2) (iii), (iv), (v)

(3) (i), (ii)

(4) (i), (ii), (iii)

62) Bioluminescence, the property of living organism to emit light, is well marked in :-

(1) Ctenophores

(2) Cnidarians

(3) Molluscs

(4) Hemichordates

63) For the following figure which one is correct ?

(A) 

(B) 

(C) 

(D) 

(1) D is animal with 3 chambered Heart

(2) C is cold blooded animal in which fertilization is external

(3) A is marine and gills are without operculum

(4) B is oviparous with external fertilization

64)

Identify the organism from the following hints :-
(A) Fresh water habitat with streamlined body
(B) Two chambered heart

(C) Air bladder present to regulate buoyancy (D) Four pair of gills with operculum

(1) *Scoliodon*

(2) *Trygon*

(3) *Myxine*

(4) *Catla*

65) Read the following statements and mark the correct answer.
Assertion (A) : The alimentary canal is short in frog.
Reason(R) : Frogs are carnivores.

(1) Both Assertion and Reason are true and reason is correct explanation of assertion.

(2) Both Assertion and Reason are true and reason is not correct explanation of assertion.

(3) Assertion is true but Reason is false

(4) Reason is true but Assertion is false.

66) Different types of excretory structures and animals are given below. Match them appropriately and mark the correct answer from among those given below :-

	Excretory organ		**Animals**
A.	Protonephridia	(i)	Prawn
B.	Nephridia	(ii)	Cockroach
C.	Malpighian tubules	(iii)	Earthworm
D.	Green gland	(iv)	Flatworm

(1) D-i, C-ii, B-iii, A-iv

(2) B-i, C-ii, A-iii, B-iv

(3) D-i, C-ii, A-iii, B-iv

(4) B-i, C-ii, A-iii, D-iv

67) Read the following four statements A-D :-
(A) All members of phylum porifera are aquatic found in marine water only
(B) Sponges are primitive multicellular animals and have cellular level of organisation
(C) In sponges fertilisation is internal and development is direct
(D) Water transport system is helpful in food gathering and respiration only not in the removal of nitrogenous wastes
How many of the above statements are incorrect ?

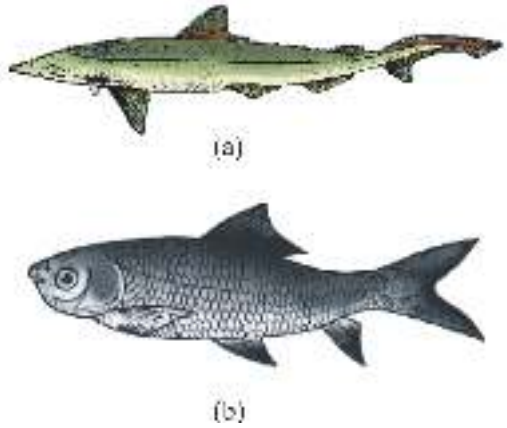
(1) Two

(2) Three

(3) Four

(4) One

68) Choose correct option about figure (a) and (b) :-



(1) In both Notochord is persistant throughout life

(2) In both, Gill slits are covered by operculum

(3) Air bladders are found in both

(4) (a) is viviporous and (b) is oviparous

69) Match column-I with column-II and select correct option about them :–

Column-I		**Column-II**	
A.	*Pristis*	i.	Sting ray
B.	*Exocoetus*	ii.	Magur
C.	*Trygon*	iii.	Saw fish
D.	*Clarias*	iv.	Flying fish

(1) A-iii B-iv C-i D-ii

(2) A-i B-ii C-iii D-iv

(3) A-ii B-iii C-iv D-i

(4) A-iv B-i C-ii D-iii

70) **Statement-I :** Any plane passing through the central axis of the body divides the organism into two identical halves, it is called Radial symmetry
Statement-II : Coelenterates, ctenophares and Echinoderms have this kind of body plan.
(A) If both Statement-I and Statement-II are True & the Statement-II is a correct explanation of the Statement-I.
(B) If both Statement-I and Statement-II are True but the Statement-II is not a correct explanation of the Statement-I.
(C) If Statement-I is True but the Statement-II is False
(D) If both Statement-I and Statement-II are false

(1) A

(2) B

(3) C

(4) D

71) Which one of the following option gives the correct categorisation of phylum accordingly to the symmetry :-

	(A) Asymmetry	**(B) Radial symmetry**	**(C) Bilateral symmetry**
(1)	Coelentreta	Ctenophora, Echinodermata (larva)	Platyhelminthes, Annelida
(2)	Porifera	Coelentrates, Arthropoda	Aschelminthes, Mollusca
(3)	Protozoa	Ctenophora , Platyhelminthes	Echinodermata, Chordata
(4)	Porifera	Coelentreta, Ctenophora	Platyhelminthes, Aschelminthes

(1) 1

(2) 2

(3) 3

(4) 4

72) Select the correct match

(1) Arthropoda : Jointed appendage with enterocoelom.

(2) Mollusca : In buccal cavity, radula is present which help in respiration.

(3) Echinodermata : Aristotle's lantern is present in sea urchin which help in mastication.

(4) Hemichorda : Closed circulatory system present.

73) In which one of the following the genus name, its two character and its class/phylum are correctly matched ?

	Genus name		**Two characters**	**Class/Phylum**
(1)	Ascaris	(a)	Body segmented	Annelida
		(b)	Males and females distinct	
(2)	Salamendra	(a)	A tympanum represents ear	Amphibia
		(b)	Fertilization is external	
(3)	Pteropus	(a)	Skin prossesses hair	Mammalia
		(b)	Oviparous	
(4)	Aurelia	(a)	Cnidoblasts	Coelenterata
		(b)	Organ level of organization	

(1) 1

(2) 2

(3) 3

(4) 4

74) Feautre are **not** related to cyclostomes is-

(1) 6-15 pairs of gill slits
(2) Mouth sucking & with jaws
(3) Shows migration for spawning
(4) Devoid of scales

75)

Consider the following statements :-
(A) Protochordates are exclusively marine
(B) In cephalopods, notochord extends from head to tail region
(C) In urochordates, notochord is present only in larval tail
(D) Cranium and vertebral column are bony in cyclostomates

Which of the above statement is/are correct ?
(1) A and C
(2) B, C and D
(3) A, B and C
(4) All of these

76) Read the following characters carefully
(a) Organ system level of organisation
(b) Sexes are separate
(c) Internal fertilization
(d) Bilateral symmetry
(e) Pseudocoelomate
How many animals given below consists of all the characters mentioned above a , b , c , d and e ?
Ascaris , Ancyclostoma , Nereis , Earthworm , Taenia, Liverfluke

(1) Two
(2) Three
(3) Four
(4) Six

77) Which one of the following categories of animals, is correctly described with no single exception in it ?

(1) All sponges are marine and have collared cells
(2) All mammals are viviparous and possess diaphragm for breathing
(3) All reptiles possess scales, have a three chambered heart and are cold blooded (poikilothermal)
(4) All bony fishes have four pairs of gills and an operculum on each side

78) Match the following columns correctly :-

Column-I	**Column-II**

(A)	These possess electric organs	(1)	*Trygon*
(B)	Animals of this class are poikilothermous	(2)	Cyclostomata
(C)	These possess poison sting	(3)	*Torpedo*
(D)	These migrate for spawning to freshwater	(4)	Chondrichthyes
		(5)	*Petromyzon*
		(6)	*Exocoetus*

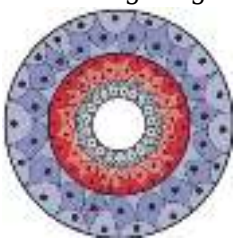
(1) A-6 B-1 C-3 D-2

(2) A-1 B-4 C-6 D-5

(3) A-3 B-4 C-1 D-5

(4) A-3 B-4 C-3 D-2

79) The figure given below shows a diagrammatic sectional view of animal. Select the correct option in which giving correct identification together with what it represent and their example ?



(1) Coelomate :- Body cavity is lined by the mesoderm, example - Annelida

(2) Pseudocoelomate :- Body cavity is not lined by mesoderm, example - Nematoda

(3) Acoelomate :- body cavity is absent, example - Platyhelminthes

(4) Coelomate :- Body cavity fills up by mesodermal pouches, example - Aschelminthes

80) **Assertion (A):** A frog never drinks water.
Reason (R): It absorb water through skin.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

81) Choose the correct function of following organs-

	A		**B**
A.	Sensory papillae	(i)	Receives sound vibrations
B.	Tympanum	(ii)	Transportation of sperms
C.	Bidders canal	(iii)	Organs of touch
D.	Sinus venosus	(iv)	Receives deoxygenated blood through venacava

(1) A - (iv), B - (ii), C - (iii), D - (i)

(2) A - (iii), B - (ii), C - (i), D - (iv)

(3) A - (ii), B - (iii), C - (iv), D - (i)

(4) A - (iii), B - (i), C - (ii), D - (iv)

82) In male frog, copulatory pad is present on the first digit of fore limbs are absent in female frog it shows-

(1) Sexes are seperate
(2) Male organs develop earlier
(3) Sexual selection
(4) Sexual dimorphis

83) A traingular structure that joins the right atrium of heart in frog is called-

(1) Sinus venosus and receives blood from pulmonary artery
(2) Conus arteriosus and receives blood from pulmonary vein
(3) Sinus venosus and receives blood from vena cava
(4) Conus arterious and receives blood from lungs

84) Find out the incorrect statement about frog

(1) In frog neck and tail are absent
(2) Eyes of frog are covered by nictitating membrane that protects them while in water
(3) Sound vibrations are received by external ear
(4) Skin is respiratory

85) Identify the correct statements :

(1) Excretory product of frog is Ammonia
(2) Mature frog female can lay 1000-1500 ova at a time
(3) Sexual dimorphism absent in frog
(4) In frog fertilization is external and takes place in water

86) During the peak summer and winter the frogs take shelter in deep burrows to protect them from extreme heat and cold. This is called as

(1) Summer sleep.(aestivation) and winter sleep.(hibernation)
(2) Summer sleep.(hibernation) and winter sleep.(aestivation)
(3) Summer sleep.(diapause) and winter sleep.(suspend)
(4) Summer sleep.(dormancy) and winter sleep.(diapause)

87) Choose the incorrect statement about vascular system of frog.

(1) Blood has plasma and cells
(2) RBCs are nucleated
(3) RBCs have haemoglobin
(4) Lymph is absent

88) Male frogs can croak than females because of having :-

(1) Larger in size
(2) Larger sound box
(3) Stronger
(4) Vocal sacs

89) Amphibia are similar to reptiles in all of the following characters except:-

(1) Tympanum represents the ear
(2) External fertilization & indirect development
(3) Dioecious and oviparous
(4) Cold blooded or poikilothermous

90) In cyclostomata fishes :-

(1) Notochord is totally replaced by vertebral column
(2) Notochord is partially replaced by vertebral is column
(3) Notochord is absent only vertebral column is present
(4) Vertebral column is present but notochord is absent

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	3	1	3	3	2	4	4	3	4	2	3	2	3	2	3	1	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	4	2	2	2	4	1	3	2	2	4	3	3	1	3	2	1	1	3
Q.	41	42	43	44	45															
A.	1	3	4	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	3	1	3	4	1	4	1	4	1	3	2	1	4	3	1	1	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	1	2	1	1	3	1	3	2	1	2	4	1	2	1	2	1	1	2
Q.	86	87	88	89	90															
A.	3	3	4	2	1															

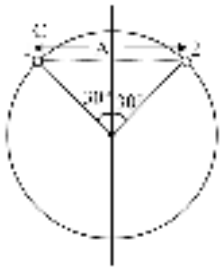
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	2	1	2	3	2	3	3	3	3	2	4	4	2	2	2	1	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	4	1	3	3	3	3	3	2	2	1	1	3	3	4	2	1	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	1	4	4	1	3	1	4	3	2	3	4	3	1	2	3	2	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	4	4	1	1	2	4	1	1	4	3	2	2	1	1	4	3	3	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	3	4	1	4	4	2	2										

SOLUTIONS

PHYSICS

1)



$\Delta\phi = 60^\circ = \left(\frac{\pi}{3}\right)$

2) Electron repeats its motion at regular intervals of time (periodic) block connected to spring performs to & fro motion at regular intervals (both) In air medium simple pendulum performs to & from motion without repeating positions at regular intervals (oscillatory)

3) $v = \omega\sqrt{A^2 - \left(\frac{2A}{3}\right)^2}$

$\Rightarrow v = \frac{\sqrt{5}A\omega}{3}$

$v_{new} = 3V = \sqrt{5}A\omega$

so the new amplitude is given by

$v_{new} = \omega\sqrt{A_{new}^2 - x^2}$

$\Rightarrow \sqrt{5}A\omega = \omega\sqrt{A_{new}^2 - \left(\frac{2A}{3}\right)^2}$

on solving $A_{new} = \frac{7A}{3}$

4)

Compare given equation with standard equation of SHM

$y = A \sin(\omega t + \varphi)$

$\varphi = \pi/3$

amplifide = A

maximum velocity = ωA

5) $\frac{d^2x}{dt^2} + 80x = 0 \Rightarrow w^2 = 80$

$\frac{4\pi^2}{T^2} = 80 \Rightarrow T = \frac{\pi}{2\sqrt{5}}$

6) $\because g_{eff} = 0 \quad T \to \infty$

7) $< KE >= \frac{1}{2} m\omega^2 A^2$

$< \cos^2(\omega t + \phi) >= \frac{1}{4} m\omega^2 A^2$

8)

NCERT BASED [Option - 4]

9)

The acceleration of the particle is maximum at the equilibrium position

10)

Let displacement equation of particle executing SHM is y = a sin ωt.

As particle travels half of the amplitude from the equilibrium position, so $y = \frac{a}{2}$

Therefore, $\frac{a}{2}$ = a sin ωt

or $\sin \omega t = \frac{1}{2} = \sin \frac{\pi}{6}$ or $\omega t = \frac{\pi}{6}$

or $t = \frac{\pi}{6\omega}$ or $t = \frac{\pi}{6\left(\frac{2\pi}{T}\right)} \left(\text{as } \omega = \frac{2\pi}{T}\right)$ or $t = \frac{T}{12}$

Hence, the particle travels half of the amplitude from the equilibrium in $\frac{T}{12}$ sec.

11)

6K	3K	K

$R_{eq} = \frac{R}{6} + \frac{R}{3} + R$

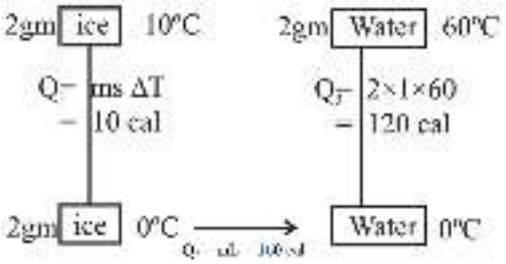
$R_{eq} = \frac{R + 2R + 6R}{6}$

$R_{eq} = \frac{9R}{6}$

$\frac{3\ell}{A.K_{eq}} = \frac{9}{6}R = \frac{9}{6} \times \frac{\ell}{KA}$

$K_{eq} = 2K$

12)



Q = 10 + 160 + 120 = 290 cal.

13) $\Delta Q_{ABC} = \Delta W_{ABC} + dU_{ABC}$

$= (\Delta W_{AB} + \Delta W_{BC}) + U_C - U_A$

$= 0 + 2P^0V^0 \ln\left(\frac{2V_0}{V_0}\right) + \frac{5}{2}[P_0(2V_0) - P_0V_0]$

$\Rightarrow \Delta Q_{ABC} = 2P^0V^0 \ln 2 + \frac{5}{2} P^0V^0$

14) W = Area enclosed between
P–V curve & V–axis

15)

Option 3 : $\frac{6}{5}$

16) If the sheet is heated then both d_1 and d_2 will increase since the thermal expansion of isotropic solid is similar to photographic enlargement.

17)

Coefficient of performance of a refrigerator is

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

Here β = 5

T_2 = –13°C = – 13 + 273 = 260 K

So, $5 = \frac{260}{T_1 - 260}$

$T_1 - 260 = \frac{260}{5}$

$T_1 - 260 = 52$

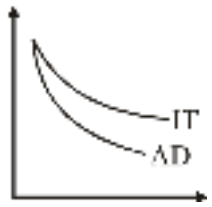
T_1 = 312 K = 312 – 273

= 39°C

18) PV^{-1} = Const

$C = C_V - \frac{R}{n-1} = \frac{3R}{2} + \frac{R}{2} = 2R$

$Q = 1 \times 2R(2T_0 - T_0) = 2RT_0$

19) 

20)

Heat gain = Heat loss
1400 × 1 (80 – 16)
= m × 540 + m × 1 × (100 – 80)
$m = \frac{1400 \times 64}{560} = 160g$

21) $P \propto T^4$
so $\frac{10}{10^5} = \frac{(427+273)^4}{T_s^4}$ $\Rightarrow T_S = 7000$ K

22)

$C = \frac{10R}{3}$
$C_V + \frac{R}{1-x} = \frac{10R}{3}$
$\frac{5R}{2} + \frac{R}{1-x} = \frac{10R}{3}$
$x = -\frac{1}{5}$
$Q = nC\Delta T = n \times \frac{10R}{3} \Delta T$
$\Rightarrow nR\Delta T = \frac{3Q}{10}$
PV^x = constant
$W = \frac{nR\Delta T}{1-x} \Rightarrow \frac{\frac{3Q}{10}}{1-\left(\frac{-1}{5}\right)}$
$W = \frac{Q}{4}$

23) $\frac{C}{5} = \frac{F-32}{9} \Rightarrow \frac{-183}{5} = \frac{F-32}{9}$
5[F–32] = –1647
F = – 329.4 + 32
≈ 297 °F

24) $\Delta t = \frac{1}{2}\alpha \, \Delta\theta \times t$

or $5 = \frac{1}{2}\alpha(\theta - 15) \times 86400$

and $-10 = \frac{1}{2}\alpha(\theta - 30) \times 86400$

Solving we get; $\frac{\theta - 30}{\theta - 15} = -2$

$\theta - 30 = -2\theta + 30$

$3\theta = 60$ or $\theta = 20°C$

25) $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = 300\left(\frac{27}{8}\right)^{\frac{5}{3}-1} = 300\left(\frac{3}{2}\right)^2$$

$$T_2 = 300 \times \frac{9}{4} = 675K$$

$\Delta T = T_2 - T_1 = 375K$

26)

Option (2) : 0^0C

27)

Option (4) : Hydrogen

28) $$W = \int PdV = \int_{V_0}^{4V_0} 2VdV = \left(V^2\right)_{V_0}^{4V_0} = 15V_0^2 = 15 \text{ units}$$

From $PV = nRT$, $2V^2 = nRT$ $2\left(V_2^2 - V_1^2\right) = nR(\Delta T) \Rightarrow nR\Delta T = 30V^{02}$

$$\Delta U = nC_V\Delta T = \frac{nR}{\gamma - 1}\Delta T = \frac{30V_0^2}{\gamma - 1} = \frac{30(1)^2}{\frac{7}{5} - 1} = \frac{30}{2}(5) = 75 \text{ units}$$

$Q = W + \Delta U = 15 + 75 = 90$ units

Molar heat capacity : $C = C_V + \frac{R}{1-x} = \frac{5}{2}R + \frac{R}{1-(-1)} = \frac{5}{2}R + \frac{R}{2} = 3R = 3 \times \frac{25}{3} = 25$ units

29)

Option (3) : 45°C

30) $$\frac{T_1 - T_2}{t} = k\left(\frac{T_1 + T_2}{2} - T_0\right)$$

$$\frac{60 - 50}{10} = k\left(\frac{60 + 50}{2} - T_0\right)$$

$$\frac{50 - 42}{10} = k\left(\frac{50 + 42}{2} - T_0\right)$$

By solving $T_0 = 10°C$

31)

As, $R = \frac{L}{KA} \Rightarrow \frac{L_1}{K_1 A} = \frac{L_2}{K_2 A}$

$\Rightarrow \frac{L_1}{L_2} = \frac{K_1}{K_2} = \frac{3}{2}$

32) $V_{rms} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + ... + V_N^2}{N}}$

$V_{rms} = \sqrt{\frac{2^2 + 3^2 + 4^2 + 5^2 + 6^2}{5}}$

$V_{rms} = 4.24$ m/sec.

33)

W = Area inside cycle = (12 – 4) × (5 – 2)

= 24 litre-atm

34) $\Delta L = \Delta L_1 + \Delta L_2$

$\alpha_{mix}(3L)\Delta T = L\alpha\Delta T + 2L(2\alpha)\Delta T$

$\alpha_{mix} = \frac{5}{3}\alpha$

35) 2kg ice at – 20°C + 5 kg water at 20°C

$Q_{gain} = Q_{lost}$

$2 \times \frac{1}{2} \times 20 + M \times 80 = 5 \times 1 \times 20$

M = 1 kg

Water = 5 + 1 = 6 kg

36)

Option (3) : 1.2×10^4 cal

37)

Option (2) : 960 K

38)

Explain:

Initial and final temperatures of the sphere and the environment are given. Find the ratio of the cooling rates.

Concept :
Stefan-Boltzmann Law:

Formula :
Cooling Rate (R): $R \propto (T^4 - T_o^4)$ where T is the temperature of the body, and T_o is the temperature of the environment

Calculation :
Ratio of Cooling Rates:

$$\frac{R'}{R} = \frac{(400)^4 - (200)^4}{(600)^4 - (200)^4} = \frac{4^4 - 2^4}{6^4 - 2^4}$$

$$\frac{R'}{R} = \frac{2^4 - 1}{3^4 - 1} = \frac{15}{80} = \frac{3}{16}$$

$$R' = \frac{3}{16} R$$

Ans (1)

39) **1. Problem Explanation:**
The rate of radiation emission of a black body follows Stefan-Boltzmann law. Given the emission rate E at $273^\circ C (546\ K)$, we need to find the emission rate at $0^\circ C (273\ K)$

2. Concept Used:
• According to Stefan-Boltzmann law, the power radiated per unit area by a black body is:

$$E \propto T^4$$

• Comparing radiation rates at two temperatures:

$$\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4$$

3. Formulas Used:
• Stefan-Boltzmann Law:

$$E_2 = E_1 \left(\frac{T_2}{T_1}\right)^4$$

4. Calculation in Minimum Steps:
• Given:

$$T_1 = 546K, \quad T_2 = 273K$$

• Using the formula:

$$\frac{E_2}{E} = \left(\frac{273}{546}\right)^4 = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

• So, $E_2 = \frac{E}{16}$

Correct Answer: (1) E / 16

40)

Option (3) : Kirchhoff's law

41) v_{rms} = same $\Rightarrow$ $T \propto M_w$

$$\frac{T_{O_2}}{T_{H_2}} = \frac{(M_w)_{O_2}}{(M_w)_{H_2}} \Rightarrow T_{H_2} = \frac{2}{32} \times 320 = 20 \text{ K}$$

42)

If Assertion is True but the Reason is False.

43) Heat at constant volume
According to FLOT : Q = DU
$\Delta U \rightarrow$ State function
So heat is also state function at constant volume.

44)

Path AB $\rightarrow$ Adiabatic process
PV^{γ} = constant.
or $TV^{\gamma-1}$ = constant

$$\frac{T_B}{T_A} = \left(\frac{V_A}{V_B}\right)^{\gamma-1} \Rightarrow T_B = 300 \times \left[\frac{0.36}{1.44}\right]^{\frac{3}{2}-1}$$

$$= 300 \times \frac{1}{2} = 150 \text{ k}$$

45)

$$W = \frac{1}{2}(P^0)(V^0)$$

$$Q_{AB} = nC_v\Delta T = \frac{3}{2}(2P^0V^0 - P^0V^0) = \frac{3P_0V_0}{2} \text{ (+ve)}$$

$$Q_{BC} = nC_p\Delta T = \frac{5}{2}[4P^0V^0 - 2P^0V^0]$$

$$= 5P_0V_0 = \text{(+ve)}$$

Q_{CA} = – ve

$$Q_{in} = Q_{AB} + Q_{BC} = \frac{13}{2}P^0V^0$$

$$\eta = \frac{W}{Q_{in}} \times 100 = \frac{100}{13}\%$$

CHEMISTRY

46)

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$
1 mol 5 mol 3 4
44 g = 132 g = 72 g

47)

W_C : WO
= 12 : 32

48) $\frac{\text{Moles}}{\text{S.C.}}$
$2P + Q \rightarrow R$
$8 \quad 5$
$\frac{8}{2} = 4 \quad \frac{5}{1} = 5$
$P \rightarrow LR$

49)

N_2 = Moles × N_A × No. of proton in 1 CH_4 molecule
$= \frac{1.6}{16} \times N_A \times 10$

50) $\frac{4 \times 1}{N_A} g$

51)

Mf = (EF)n
$n = \frac{78}{13} = 6$
Hf = $(CH)_6$

52) $0.334 = \frac{56 \times x}{67200} \times 100$
x = 4

53)

No. of Atoms = Moles × N_A × Atomicity
$= \frac{6.4}{64} \times N_A \times 3$

$H_2 + Cl_2 \longrightarrow 2HCl$

22.4 11.2 L 2 mole

1 mole 0.5 mole

(L.R.)

As 1 mole Cl_2 – 2 moles of HCl

So, 0.5 mole of Cl_2 – 1 mole of HCl

54) i.e., 1 mole of HCl (g) is formed.

$$\text{No. of mole} = \frac{V(\ell) \text{ occupied by a gas at STP}}{22.7}$$

55) or, $\text{vol of gas at STP} = 2 \times 22.7 = 45.4\ \text{L}$

56) $d = \frac{M}{V}$ $M = d \times V = 1 \times 18 = 18g.$ $n = \frac{18}{18} = 1$

$n = \frac{0.18}{18} = 0.01$

$n = \frac{10^{-3}}{18}$

$n = \frac{0.00224}{22.4} = 10^{-4}$ $w = 10 \times 10^{-4} g$

57) $\frac{\frac{W_{H_2}}{M_{H_2}}}{\frac{W_{O_2}}{M_{O_2}}} = \frac{\frac{1}{2}}{\frac{4}{32}} = 4 : 1$

58) $8 = \frac{32 \times 1}{MM} \times 100$

MM = 400

59) $d = \frac{M}{22.4} = \frac{32}{22.4} = 1.43$

60) $x = \frac{140\times}{28} = 5 \times 2 = 10$

	Mass%	atomic ratio	simplest ratio
M	60	$\frac{60}{24}=2.5$	$\frac{2.5}{2.5}=1$
O	40	$\frac{40}{16}=2.5$	$\frac{2.5}{2.5}=1$

61) $\boxed{E.F=MO}$

62)

wt. of 1 L of hydrocarbon = wt. of 1 L of CO_2
 wt of 22.4 L of hydrocarbon = wt. of 22.4 L of CO_2
(mol. wt of CO_2) = 44 g
Ans → C_3H_8 have mol. wt = 44

63) $\frac{n_{Al}}{2}=\frac{n_{O_2}}{\frac{3}{2}}$

$$\frac{18}{27\times 2}=\frac{w}{32\times\frac{3}{2}}$$

w = 16 g.

64)

$CaCO_3 \rightarrow CaO + CO_2$

$$\frac{25}{100}=\frac{1}{4} \qquad \frac{1}{4}=\frac{V}{22.4}$$

V = 5.6 lit.

65)

$CO_2(g) + C(g) \rightarrow 2CO_2(g)$

V 2V

26 cm^3 $2 \times 26 = 52\ cm^3$

66) 1.8 atm

67)

2 mole of Al reacts with 3/2 mole of O_2 to produce 1 mole of Al_2O_3

68) **Explain Question:** We are asked whether the equivalent mass of an element is variable and why.

Concept: Equivalent mass=Atomic mass / Valency
Solution: Valency of an element may vary in different compounds, so equivalent mass also changes.
Assertion: True (Equivalent mass is variable).
Reason: True (It depends on valency). Reason correctly explains the Assertion.
Final Answer: Option 1

69) $\underset{20.8}{BaCl_2} + \underset{9.8}{H_2SO_4} \rightarrow \underset{x}{BaSO_4} + \underset{7.3}{2HCl}$
x = (20.8 + 9.8) - 2.3
= 23.3

70) $K_2CO_3 \quad CO_2$
POAC on C atom
$n_{K_2CO_3} = \frac{138}{138} = n_C = \frac{12}{12} = 1$

71)

CO, CO_2

72)

No. of atoms = moles × N_A × atomic
= 1.5 × N_A × 6
= 9 N_A

73) $V.D = \frac{O_3}{CH_4} = \frac{48}{16} = 3$

74) $M \times \frac{20}{100} = 7$

75) $MF \cdot C_6H_{12}O_6$
$EF = CH_2O$

76)

1. Question Explanation
We are given two reactions involving the decomposition of two oxides, MO and MO_2. We need to determine which oxide is more stable based on their equilibrium constants.
2. Given Data

A. Reaction 1: $2MO(g) \rightleftharpoons M_2(g) + O_2(g)$; $K_1 = 1 \times 10^{-10}$
B. Reaction 2: $2MO_2(g) \rightleftharpoons M_2(g) + 2O_2(g)$; $K_2 = 4 \times 10^{6}$

3. Concept

A. **Equilibrium Constant (K):** A higher value of K indicates that the reaction favors the formation of products at equilibrium. In this case, a higher K suggests that the oxide is less stable and more likely to decompose.

4. Mathematical Calculation

Not applicable for this qualitative analysis.

5. Final Answer

Based on the given equilibrium constants, MO is more stable then MO_2.

A. Reaction 1 has a significantly higher K value (1×10^{-10}) compared to Reaction 2 (4×10^{6}). This indicates that MO decomposes to M_2 and O_2 much more readily than MO_2.

Therefore, MO_2 is more stable.

77)

Question Explanation : We need to compare K_p and K_c. Relation is
$K_p = K_c (RT)^{\Delta n}$
If $\Delta n < 0 \rightarrow K_p < K_c$.
Concept:- This question is based on Δn = moles of gaseous products - moles of gaseous reactants.
Solution :-
1. $N_2O_3(g) \rightleftharpoons NO_2 + NO$
$\Delta n = (1 + 1) - 1 = +1 \rightarrow K_p > K_c$
2. $2H_2 + O_2 \rightleftharpoons 2H_2O$
$\Delta n = 2 - (2 + 1) = -1 \rightarrow K_p < K_c$
3. $H_2 + I_2 \rightleftharpoons 2HI$
$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$
4. $2SO_3 \rightleftharpoons 2SO_2 + O_2$
$\Delta n = (2 + 1) - 2 = +1 \rightarrow K_p > K_c$
Final Answer : Option (2)
$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

78) $K_c = \frac{1}{[O_2]^5}$

79)

Option (1) : $K = 10^3$

80)

Explanation Question:
How does temperature affect the equilibrium constant.
Concept:
This question is based on Le-chatelier principle and the van't hoff equation explain this.
Solution:

For exothermic reaction (ΔH < O) increasing the temp. shifts the equilibrium to the left, favoring reactants. This causes the equilibrium constant (Keq) to decrease for endothermic reactions (ΔH > O), increasing the temp shifts the equilibrium to the right, favoring products. This causes the equilibrium constant to increase.

Final Answer: (2)

81) $$K_C = \frac{1}{(K_1)^2} = \frac{1}{25}$$

82) $Pv_5 \rightleftharpoons Pv_3 + u_1$

4-0.5×4 (0.5×4) (0.5×4)

2 2 2

$n_T = 2 + 2 + 2 = 6$

83) $$\log \frac{K_2}{K_1} = \frac{-\Delta H}{2.303R} \left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

For endothermic → ΔH = +ve

and $T_2 > T_1$

So, $\frac{1}{T_2} - \frac{1}{T_1} \Rightarrow \frac{T_1 - T_2}{T_2 T_1} < 0$ $(T_1 - T_2 < 0)$

$\log \frac{K_2}{K_1} > 0$ $\Rightarrow$ $\log K_2 > \log K_1$

84) $$Q_C = \frac{\left(\frac{6}{2}\right)^2}{\left(\frac{2}{2}\right)\left(\frac{4}{2}\right)^3} = \frac{9}{8}$$

$Q_c < K_c$ so reaction will proceed in forward direction.

85) **Question Explanation : $N_2 + 3H_2 \rightleftharpoons 2NH_3$**

Given:

- Inert gas (angon) is added
- Pressure is kept constant
- The system is gaseous and dynamic equilibrium exists

Concept:

affecting factor of Equilibrium

Explanation :

At constant pressure, adding an inert gas increases the total volume to keep pressure constant.

This leads to a decrease in partial pressures of the reacting gases

The system will now shift to the tide with more moles of gas to increase pressure again In the given reaction

- Reactants: 1 mol N_2,+ 3 mol H_2 = 4 mols
- Products 2 mol NH_3

So, since the reactant side has more gas molecules, equilibrium will shift towards the left

(backward) to increase the total number of gas molecules

Final Answer: 2. Shift the equilibrium in backward direction
At constant pressure, adding argon cures the equilibrium to shift backward.

86)

Addition of inert gas at constant volume has no effect on equilibrium.

87) $$\alpha = \frac{n_1}{Dry}\left(\frac{D_T - D_0}{D_0}\right) \times 100$$
$$\alpha = \frac{1}{1}\left(\frac{46-30}{30}\right) \times 100 = 53.3\%$$

88)

Question Explanation
Relate K_1, K_2, K_3 for the given equilibria.

Concept
Use equilibrium expressions and substitution method.

Solution

I: $A + 2B \rightleftharpoons C,\ K_1 = \frac{[C]}{[A][B]^2} \Rightarrow [C] = K_1[A][B]^2.$

II: $C + D \rightleftharpoons 3A,\ K_2 = \frac{[A]^3}{[C][D]} \Rightarrow [D] = \frac{[A]^3}{K_2[C]}.$

III: $6B + D \rightleftharpoons 2C,\ K_3 = \frac{[C]^2}{[B]^6[D]}.$

Substitute [D]:

$$K_3 = \frac{[C]^2}{[B]^6} \cdot \frac{K_2[C]}{[A]^3} = K_2\frac{[C]^3}{[A]^3[B]^6}.$$

Now use $[C] = K_1[A][B]^2$:

$$[C]^3 = K_1^3[A]^3[B]^6.$$

So, $K_3 = K_2 \cdot \frac{K_1^3[A]^3[B]^6}{[A]^3[B]^6} = K_1^3K_2.$

Final Answer

$K_3 = K_1^3 \times K_2$ – Option 4.

89)

Melting of ice increases on increasing pressure and temperature.

90) Both A and R are true and R is the correct explanation of A.

BIOLOGY

91) NCERT Pg. # 17

92)

NCERT XI Pg. # 22,23

93)

Explanation:
Which on of the following structure is similar in E. coli and chlamydomonas.

Concept:
This question is based on E coli and chlamydomonas is prokaryote eukaryote.

Solution:
The structure similar in E.coli and chlamydomonas is cell membrane.

Final Answer:
Option (2)

94)

Question Explanation:
Choose correct match from the given options

Concept: Chrysophytes: cell wall is made up of silice which indestructible.

Solution:
Diatoms and desmids are known as chrysophytes where cell wall is made up of silica.
Here cell wall is indestructible.
So the correct option is 1. Chrysophytes.

95) NCERT XI Page No. # 23 & 24

96) NCERT - 128, 20, 22

97) NCERT XI Pg. # 20,21
(2.2.1, 2.2.2, 2.2.3, 2.2.4)

98) NCERT XI Pg. # 17,18

(Introduction)

99) NCERT XI Pg. # 20,21
(2.2.1, 2.2.2, 2.2.3, 2.2.4)

100) NCERT XI Pg. # 23
(2.3)
Ascospore, basidiospore and oospores

101) NCERT XI Pg. 20, 21, (2.2.1, 2.2.2, 2.2.3, 2.2.4)

102) NCERT XI, Page # 17

103) NCERT-XI, Pg. # 28, Exercise Que. # 1 & answer on 18

104) NCERT XI Pg.# 20

105) NCERT, Page No. # 18

106)

All moreran have flagella

107) NCERT Pg.# 21

108) Bacteria Kingdom
Blue-green algae – Prokaryotes
Nitrogen-fixing bacteria – Prokaryotes
Methanogenic bacteria – Prokaryotes

109) NCERT Page # 23

110) NCERT-Page # 14

111) NCERT-XI, Pg. # 17, 21

112) NCERT Pg. # 18, 19

113) NCERT Pg. # 26

114) NCERT-XI, Pg # 26,27

115) NCERT XI, Pg. # 22

116) NCERT Pg # 20

117) **Explanation :-** The question asks to identify the animal-like feature of slime moulds from the provided options.
Concept :- This question is based on Kingdom Protista - Slime moulds
Solution :- Absence of cell wall in plasmodium - This is an animal-like feature. In the plasmodial stage, slime moulds lack a cell wall, allowing them to move and engulf food particles, similar to animals.
Answer :- 3

118) **Question Explanation:** Classifying organisms based on evolution.
Concept: Phylogenetic Classification.
Solution: The **Phylogenetic system** classifies organisms based on their **evolutionary relationships** and common ancestry. It assumes that a taxon's members have a common ancestor, which is a key difference from other systems that use superficial or a limited number of traits.
Final Answer: (3) Phylogenetic system

119) NCERT XI, Pg. # 18

120) **Correct Answer:** Option 2: A-II, P ; B-I, P, Q, R ; C-III, S, P ; D-I, IV, R
Explanation:

A. Monera (A) → II (Nuclear membrane absent) ; P (Parasitic and saprophytic)
    A. Bacteria (Monera) are prokaryotic, meaning they lack a nuclear membrane.
    B. They can be parasitic (causing diseases) or saprophytic (feeding on dead matter).
B. Protista (B) → I (Cellulosic cell wall) ; P (Parasitic and saprophytic) ; Q (Holozoic) ; R (Photosynthetic)
    A. Protists are unicellular eukaryotes with a nuclear membrane.
    B. Some have cellulosic cell walls (e.g., algae), while others are holozoic (ingest food like Amoeba).
    C. They can be photosynthetic (e.g., Euglena) or parasitic (e.g., Plasmodium).
C. Fungi (C) → III (May form loose tissue) ; S (Cell wall made of chitin) ; P (Parasitic and saprophytic)
    A. Fungi have chitin in their cell wall.
    B. They grow as loosely organized mycelial networks.
    C. Most fungi are saprophytic (decomposers), but some are parasitic (causing diseases).
D. Plantae (D) → I (Cellulosic cell wall) ; IV (Tissue/organ grade of organization) ; R (Photosynthetic)

A. Plants have a cell wall made of cellulose.
B. They show tissue and organ-level organization (roots, stems, leaves).
C. They are autotrophic (photosynthetic).

121)

Solution:

The correct answer is:2. Protista
According to Whittaker's 5 kingdom classification, organisms like Chlorella (alga), Chlamydomonas (alga), Paramoecium, and Amoeba (protozoans) are all unicellular eukaryotes and should be classified under Protista, which is a separate kingdom for such organisms.

122) NCERT XI(E)/(H) Page No. # 10 (1.37) last para

123) NCERT XI(E)/(H) Page No. # 11, Table-1.1

124) NCERT XI Pg # 11 Table

125)

Module No.1, Pg # 11

126) All the statements regarding nomenclature are correct. Nomenclature is giving distinct scientific names to various structures including living organisms for their identification. It is a set of rules used for forming the names or terms in a particular field of arts or sciences. Nomenclature is only possible when the organism is described correctly and we know to what organisms the name is attached to (called identification)

127)

Ans. (2) & (3)

128) NCERT Pg. # 10

129) NCERT Pg # 11

130) NCERT-XI, Pg. # 08

131) NCERT-XI, Pg. # 09

132) NCERT-XI, Pg. # 11

133)

NCERT page no. 21

134) NCERT-XI, Page No. 8

135) NCERT, Pg. # 7

136) **Question Explanation:** Identify the feature that is not characteristic of chordates.
Concept: Chordata - Fundamental features
Solution: The central nervous system in chordates is **dorsal, hollow, and single**. A **ventral nerve cord** is characteristic of non-chordates.
Final Answer: 1

137) NCERT (XI) Pg# 56 Para: 4.2.11.2

138) **Question Explanation:** The characteristic feature of birds which makes them different from reptiles is:

Concept : This question is based on comparison of reptiles and aves

Explanation : The homoeothermic (warm-blooded) nature is a unique feature that differentiates birds from reptiles.

Final answer : Homoeothermic nature

139)

Explain Question:
Asking to match the correct.

Concept:
The concept based on Intra porifera to aschelminthes

Solution:

Gorgonia- (Sea fan)-2

Ctenoplana- (Comb jelly)-3

Taenia- (Tapeworm)-4

Euspongia- (Bath sponge)-1

Final answer: option (4)

140) **Question Explanation:** Germ layer terminology

Concept: Number of germinal layers
Solution: Diploblastic and triploblastic describe the number of germinal layers (two or three) present during embryonic development in animals.
Final Answer: The number of germinal layers during embryonic development

141)

XI^{Th}_NCERT Pg No :- 46

142)

NCERT-XI; Page No.#47,49,50

143) **Question Explanation:** Identifying which organism lacks an open circulatory system.

Concept: Circulatory System Types

Solution: An **open circulatory system** is not present in **Annelida** (e.g., earthworms and Nereis). Annelids have a **closed circulatory system**, where blood is contained within a network of blood vessels. In an open system, which is found in arthropods and most molluscs, blood (hemolymph) is pumped into an open body cavity called a hemocoel.

Final Answer: Annelida (Earthworm, Nereis)

144)

Question Explanation: Identifying correctly matched common and scientific names.
Concept: Animal nomenclature.
Solution: Aplysia is a sea hare. Fasciola is a liver fluke. Physalia is a Portuguese man of war. Antedon is a sea lily. The correct match is option 3.
Final Answer: 3

145) **Explain Question :** Metagenesis means
Concept : This question is based on metagenesis
Explanation : In Obelia and other cnidarians, polyps reproduce asexually to form medusae, while medusae reproduce sexually to form polyps.
Final answer : 1: Polyp produce medusae asexually and medusae produce polyp sexually

146)

Pg. No. 57 NCERT 2022 - 2023 Edition

147)

NCERT-XI; Page No.# 50

148)

Ncert page 48 to 51.

149) **Question Explanation:** Identify the characteristics of class Reptilia

Concept: Reptilia characteristics: skin and scales

Solution: Reptiles are characterized by a body covered with dry and cornified skin with epidermal scales or scutes, an adaptation for terrestrial life.
Other options describe amphibians, bony fish, and cartilaginous fish.

Final Answer: Option-4

150) **Question Explanation:** Identify the parts labeled in the schematic of a chordate.
Concept: Chordata - Fundamental features
Solution: The correct identification is: A - **Pharyngeal gill slits** (openings in the pharynx). B - **Nerve Cord** (dorsal and hollow). C - **Post Anal tail** (extends beyond the anus). D - **Notochord** (mesodermal rod beneath nerve cord).
Final Answer: 3

151) NCERT (XI) Pg. # 58

152)

Question Explanation: Identify the Organisms that produce light.
Concept: Bioluminescence.
Solution: Bioluminescence, the emission of light by a living organism, is a prominent characteristic of **ctenophores**. They are exclusively marine animals, and nearly all species are bioluminescent.
Final Answer: option (1) Ctenophores

153)

NCERT Pg. # 57, 58, 59

fig.4.20(a), 4.21(b), 4.23(b), 4.22(b)

154)

NCERT-XI, Pg#57

155) NEW NCERT XI Page no. 81 Para 7.2.2

156)

NCERT-XI; Page No. # 43,44

157) NCERT Pg. # 49, Para-4.2.1

158)

NCERT-XI; Page No. # 47, 48

159) NCERT Pg. # 57, Para-4.2.11.2, 4.2.11.3

160)

NCERT-XI; Page No. # 38

161)

NCERT-XI; Page No. # 38

162)

NCERT-XI; Page No. # 45

163)

NCERT-XI; Page No. # 48

164) NCERT Pg. # 56

165)

Question Explanation: Identifying correct statements about chordate subphyla.
Concept: Chordate characteristics.
Solution: Statement (A) and (C) are correct. Cephalopods (not cephalochordates) lack notochords. Cyclostomes have a cartilaginous skeleton, not a bony one.
Final Answer: 1

166)

NCERT-XI; Page No. # 43

167)

NCERT-XI; Page No. # 48

168)

NCERT-XI; Page No. # 48

169) NCERT XI (H) Pg. # 48, Fig. 4.3

170) NCERT pg. # 116

171) NCERT pg. # 19

172) NCERT pg. # 116

173) NCERT pg. # 118

174) NCERT pg. # 119

175) NCERT, Pg # 83, 84

176) NCERT - XI, Pg. # 116

177)

NCERT-XI; Page No. # 82

178)

NCERT-XI; Page No. # 81

179)

NCERT-XI; Page No. # 80

180)

NCERT-XI; Page No. # 47